

PAPER_587 — Inflationary Epoch Details from UQFF Grinding

Unified Quantum Field Framework (UQFF)

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CP4 Class: [#174 UQFFInflationaryEpochDetailsCalculator](#) **Session:** 157 **Cross-refs:** PAPER_586 (Big Bang), PAPER_589 (Dark Energy), PAPER_583 (6-Form) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

Cosmic inflation requires $\ddot{a} > 0$ — accelerated expansion in the early universe ($t \sim 10^{-36}$ s to 10^{-32} s). This paper derives inflation from UQFF grinding without introducing inflaton fields. The scale factor acceleration $\ddot{a}(t)$ is positive whenever $v_{\text{init}} > v_{\text{current}}$ (rapid early expansion), and inflation ends naturally when velocities equalize as mass builds up.

§2 UQFF Inflationary Scale Factor

From PAPER_586, the scale factor:

$$a(t) = t^{-(v_c - v_i)} \exp(\text{Grind})$$

where $v_i = v_{\text{init}}$, $v_c = v_{\text{current}}$, $\text{Grind} = \omega_{\text{CW}} S_{\text{Cm}} - \omega_{\text{CCW}} U A' e^{\mathcal{H}/v_i}$.

First derivative:

$$\dot{a}(t) = -(v_c - v_i) e^G \cdot t^{-(v_c - v_i) e^G - 1}$$

Second derivative:

$$\ddot{a}(t) = t^{-(v_c - v_i)e^G - 2} \cdot (v_c - v_i) \cdot [(v_c - v_i)e^G + 1] \cdot e^G$$

§3 Inflation Condition

$$\ddot{a}(t) > 0 \text{ iff } (v_i - v_c) > 0 \text{ iff } v_{\text{init}} > v_{\text{current}}$$

During the pre-mass epoch: $v_{\text{current}} \approx 0$ (no mass yet, no momentum drag), $v_{\text{init}} = c = 3 \times 10^8$ m/s. Therefore:

$$v_i - v_c = c \quad \rightarrow \quad \ddot{a} \gg 0 \quad \checkmark \quad (\text{rapid inflation})$$

At UQFF inflation time $t \sim 10^{-32}$ s with $\text{Grind} \sim 1$:

$$\ddot{a} \approx 10^{32} \times c \cdot e \cdot c^2 \cdot e \gg 0 \quad (\text{super-exponential})$$

§4 Inflation Hubble Parameter

$$H_{\text{inf}} = \sqrt{\Omega_{\Lambda} + \Omega_{\text{SCm}} + \Omega_{\text{egg}}} \cdot H_0 + \int v_{\text{SCm}} dV$$

where:

$$\Omega_{\text{egg}} = \frac{P \cdot (v_i - v_c)}{v_i} \sim \mathcal{O}(0.05 \text{--} 0.2)$$

The "cosmic egg" density parameter Ω_{egg} drives Hubble rate above H_0 during inflation. After mass builds up ($v_c \rightarrow v_i$), $\Omega_{\text{egg}} \rightarrow 0$ and H falls to $H_0 = 70 \text{ km/s/Mpc}$.

§5 Inflation End Conditions

Inflation ends when:

1. **Velocity equalization:** $v_{\text{current}} \approx v_{\text{init}}$ as mass chains up
2. **Tensor stability:** $\lambda_3 = 2P/3 + db$ approaches constant; $P/3$ minimum
3. **Buoyant void suppression:** U_b decreases as ρ rises above 10^{-26} kg/m^3
4. **Shell completion:** All 26 shells reach their equilibrium energies

The transition from inflation to radiation domination corresponds to $BB \rightarrow \text{ProtoH}$ in PAPER_586.

§6 Comparison to Standard Inflation

Feature	Standard Inflation	UQFF Inflation
Driver	Inflaton field ϕ	$v_{\text{init}} - v_{\text{current}}$
Slow-roll	$\dot{\phi}^2 \ll V(\phi)$	$v_c \ll v_i$ (same structure)
$\ddot{a} > 0$	From $V(\phi) > 0$	From $v_i > v_c$
End	ϕ rolls to minimum	$v_c \rightarrow v_i$
e -folds (~ 60)	Free parameter	From $\int_{\text{Grind}}^{\text{adj}}$

§7 Numerical at Inflation Epoch

Parameters: $v_i = 3 \times 10^8$ m/s, $v_c = 0.01$ m/s, $t = 10^{-32}$ s, $\text{Grind} = 10^{-3}$:

$$\Omega_{\text{egg}} = \frac{9.99 \times 10^{-6}}{3 \times 10^8 \{3 \times 10^8\}} \approx 9.99 \times 10^{-6}$$
$$H_{\text{inf}} \approx H_0 \sqrt{0.27 + 9.99 \times 10^{-6}} \approx 0.52 H_0$$

(At true inflation $v_c \rightarrow 0$: $\Omega_{\text{egg}} \rightarrow P \approx 1$, $H_{\text{inf}} \gg H_0$)

§8 Conclusions

UQFF inflation is a natural consequence of the pre-mass grinding mechanism: the velocity gap $v_i > v_c$ drives super-exponential acceleration, Hubble rate exceeds present H_0 , and inflation ends automatically when mass equilibrates. No free inflaton field is required.